Themistoklis Nikas

Ph.D. Student in Computer Science
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Education

Boston University Boston, MA, USA
Ph.D. in Computer Science; Advisor: Prof. Aaron Mueller *Sep 2023 – May 2028 (expected)*

- Mechanistic interpretability of LLMs, focusing on confidence estimation and calibration. Current work uses linear representations of ambiguity to improve how well models' confidence reflects their correctness.

University of Ioannina Ioannina, Greece
B.Sc. & M.Sc. in Computer Science & Engineering (5-year joint degree; 300 ECTS) *Mar 2017 – Mar 2022*

- Diploma Thesis: *Fairness in Link Analysis Algorithms* (Apr 2021 – Mar 2022): achieving fairness in PageRank with minimal network modifications, with theoretical and empirical analysis; led to a publication.

Publications

Improving Confidence Estimation in Language Models with Linear Representations of Ambiguity
***Nikas, T.**, Subramani, N., Mueller, A. (Actionable Interpretability @ COLM 2026); under submission*

Minimizing the Cost of PageRank Fairness
*Tzimas, S., **Nikas, T.**, Pitoura, E., Tsaparas, P. (WebGraph @ WSDM 2026)*

Disentangling Text and Math in Word Problems: Evidence for the Bidimensional Structure of Large Language Models' Reasoning
*Calais, P., Franco, G., Tang, Z., **Nikas, T.**, Meira Jr., W., Terzi, E., Crovella, M. (Findings of ACL 2025)*

Beyond Accuracy: Understanding the Performance of LLMs on Exams Designed for Humans
*Calais, P., Franco, G., **Nikas, T.**, Tang, Z., Crovella, M., Meira Jr., W., Terzi, E. (Preprint, 2025)*

Research Experience

Research Associate Feb 2023 – Aug 2023
University of Ioannina & Pfizer CDI Collaboration *Thessaloniki, Greece*

- Built ML/NLP tools to assess health misinformation on Reddit, collecting subreddits on hoaxes and rumors and verifying claims with fact-checking tools and databases.
- Clustered posts with text embeddings and analyzed them via graph analysis, feature extraction, and statistics.

Junior Researcher May 2022 – Jan 2023
Unit of Medical Technology & Intelligent Information Systems (MedLab) *Ioannina, Greece*

- Researched probabilistic and ML methods for synthetic omics generation and virtual population analysis.
- Investigated the relationship between cardiovascular diseases and depression using ML classification.

Teaching Fellow

Boston University Jan 2024 – Dec 2025

- Introduction to Analysis of Algorithms (Spring 2024, Fall 2024, Spring 2025, Fall 2025). Led discussion sections, held office hours, and helped design assignments and exams.

University of Ioannina Feb 2023 – Jun 2023

- Object-Oriented Programming; supported students through labs, grading, and Q&A.

Service & Activities

Reviewer: Actionable Interpretability @ COLM 2026
Co-organizer: New England Mechanistic Interpretability Workshop (NEMI 2026)
Talk: *Minimizing the Cost of PageRank Fairness*, WebGraph @ WSDM 2026

Skills, Languages & Volunteering

Programming & Tools: Python, PyTorch, Hugging Face, TransformerLens, nnsight, scikit-learn, Git, LaTeX
Languages: English (fluent), Greek (native)
Volunteering (TEDxMaviliSquare, 2021–2023): Oversaw speaker recruitment, content direction, and workshops.